

PAVLO SKRYPKA

Mechanical Design Engineer | AutoCAD · ANSYS Workbench · structural and thermal FEA

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PROFILE

Engineering assistant on a gyroscope development project. I produce manufacturing documentation in AutoCAD daily and run structural and thermal analysis in ANSYS Workbench. I verify results rather than accept them: I can tell a numerical mesh artefact from a real stress. Fourth-year student at Igor Sikorsky Kyiv Polytechnic Institute, available full-time now.

EXPERIENCE

Engineering Assistant

July 2025 — present

Instrument manufacturing, gyroscope development — Kyiv

- Manufacturing documentation in AutoCAD: machined parts, weldments, assembly drawings and specifications to state standards.
- Structural and thermal analysis in ANSYS Workbench (Static Structural, Steady-State Thermal, Explicit Dynamics): meshing, boundary conditions, safety factor assessment.
- Verification of results: FEA cross-checked against closed-form solutions from strength of materials; mesh convergence confirmed before conclusions are issued.
- Specification automation: initiated and rolled out a tool that cut specification preparation from hours to seconds and removed manual transcription errors.

SELECTED ANALYSIS

Coupled thermal-structural analysis of a cast iron pump cover

ANSYS Workbench

- Steady-State Thermal + Static Structural. Before reading results, verified the energy balance from reaction probes on all boundary conditions — closure to 0.001% of the heat input.
- Peak stress at constrained hole edges identified as a numerical singularity and excluded from the assessment; stress in the flange body 115.3 MPa, margin 2.1 against ultimate strength.

Static structural analysis of an aluminium enclosure

ANSYS Workbench

- 0.9 MPa external pressure; coarse mesh compared against a refined 0.95 mm mesh. Minimum safety factor 1.13 — recommended holding mass production release and increasing wall thickness and fillet radii.

High-speed impact on a fluid-filled vessel (FSI)

ANSYS Explicit Dynamics

EDUCATION

Igor Sikorsky Kyiv Polytechnic Institute

bachelor's

Institute of Aerospace Technologies — fourth year

SKILLS

CAD: AutoCAD — confident, daily production work. SolidWorks — working knowledge, currently deepening.

CAE: ANSYS Workbench: Static Structural, Steady-State Thermal, Explicit Dynamics. Meshing and convergence control, singularity assessment.

Analysis: Strength of materials, analytical verification of FEA results, reading datasheets and material references.

Also: Python — basic. FPGA (Xilinx Vivado, VHDL) — introductory. Currently learning C and embedded systems.

Languages: Ukrainian — native. English — B1, reads technical documentation comfortably.